

MoSAIC: Motion- and Supervision-Aware Inference for Myocardial Scar and Edema Segmentation from Multi-Sequence and Cine CMR

Anonymous Author(s)

Institution masked for double-blind review

Abstract. Segmenting myocardial scar and edema from cardiac magnetic resonance (CMR) quantifies infarct burden and salvageable myocardium, while realistic multi-center data require models that use heterogeneous evidence. Sequence availability and label spaces vary across centers, so supervision should follow the annotated class set of each case. Cine CMR adds complementary motion evidence: scar appears mainly as regional wall-motion abnormality. We propose MoSAIC (Motion- and Supervision-Aware Inference for Myocardial Scar and Edema Segmentation), a coarse-to-fine, heterogeneity-aware, anatomy-conditioned cascade for multi-sequence and cine CMR. A coarse stage first estimates cardiac anatomy and defines a spatial prior. The fine stage then combines a multi-sequence scar/anatomy decoder, a dedicated edema-zone expert and a cine motion scar head. Presence-gated encoding with a supervision-masked objective ensures that each case supervises only its annotated classes, allowing heterogeneous cohorts to be pooled with case-aware supervision. The cine pathology head uses a dense end-diastole-anchored motion field as kinematic evidence, and the edema expert predicts the injury-zone label used for scar/edema fusion. Official validation on CARE-2026 CARE-Myocardium shows best MoSAIC Dice entries of 0.6965 for MyoPS scar, 0.5983 for MyoPS edema and 0.2069 for CineMyoPS scar. Internal five-fold evaluation reports Dice of 0.668 for scar and 0.645 for lesion union on multi-sequence CMR, and 0.457 for cine-only scar.

Keywords: Myocardial pathology segmentation · Multi-sequence CMR · Cine CMR · Motion-derived evidence · Partial supervision.

1 Introduction

After acute myocardial infarction, scar marks irreversibly necrotic myocardium while edema marks salvageable injured tissue. Their relation guides reperfusion assessment [6,9], and CMR resolves them through complementary contrasts: LGE highlights scar [11], T2-weighted imaging highlights edema [1], and bSSFP/C0 supplies anatomy. Automating their joint delineation is the goal of the MyoPS benchmark family [12,28].

The CARE-2026 setting emphasizes realistic heterogeneity in both image evidence and annotation protocols. In multi-sequence CMR, training cases may contain LGE, LGE+C0 or the full LGE+C0+T2 combination; 80 of 220 cases annotate edema, and centers provide different anatomical label sets. Effective training therefore requires the objective to respect the class set that each center annotated. In cine-only CMR, scar is represented primarily through regional wall-motion abnormality [26,3], making motion a clinically meaningful signal for contrast-free pathology inference.

We propose a coarse to fine heterogeneity-aware framework that models both motion evidence and supervision availability (Fig. 1). The coarse stage predicts cardiac anatomy, extracts a region of interest and builds a spatial prior. Inside this prior, Stage 2a decodes multi-sequence scar with auxiliary anatomy consistency, Stage 2b predicts the challenge injury zone (edema $\cup$ scar), and Stage 2c derives cine scar from ED-anchored motion and warped anatomy. A supervision-masked loss ensures that each case supervises only its annotated classes, enabling heterogeneous centers to be pooled while preserving their annotation semantics.

Our contributions are a coarse to fine, heterogeneity-aware cascade that links anatomy localization with scar/anatomy and edema-zone decoding; a presence-gated, supervision-masked recipe for ragged multi-center labels; a motion-driven cine scar head; and a five-fold CARE-2026 evaluation with organizer-scored validation results.

2 Related Work

The MyoPS challenges established joint scar/edema segmentation on paired LGE, T2 and bSSFP data [28,12]. Subsequent methods improved flexible sequence use [18] and joint alignment-segmentation [4], with a primary focus on unified annotation spaces. Generic U-Net and nnU-Net pipelines provide strong segmentation backbones [19,10]; MyoPS variants have also used coarse cardiac localization before fine scar/edema decoding [25,24]. The proposed framework follows this anatomy-prior intuition and extends it to variable labels, variable sequence availability and cine motion evidence.

Missing-modality methods such as HeMIS handle variable inputs through pooled or imputed latent statistics [8,5], while partial-label losses preserve the semantics of classes outside a case’s annotation set [22,27,7]. The proposed framework combines the two requirements in one mechanism: modality presence gates and a per-class supervision mask in the loss normalizer.

Contrast-free cine CMR offers a complementary route to pathology inference. Prior work has used cine appearance [26], strain [21,16], or joint motion-and-segmentation learning [17]; CineMyoPS aggregates motion and anatomy branches across the cardiac cycle [3]. The proposed framework follows the same clinical motivation and feeds the scar head motion-derived evidence, emphasizing the kinematic signal most directly tied to cine CMR.

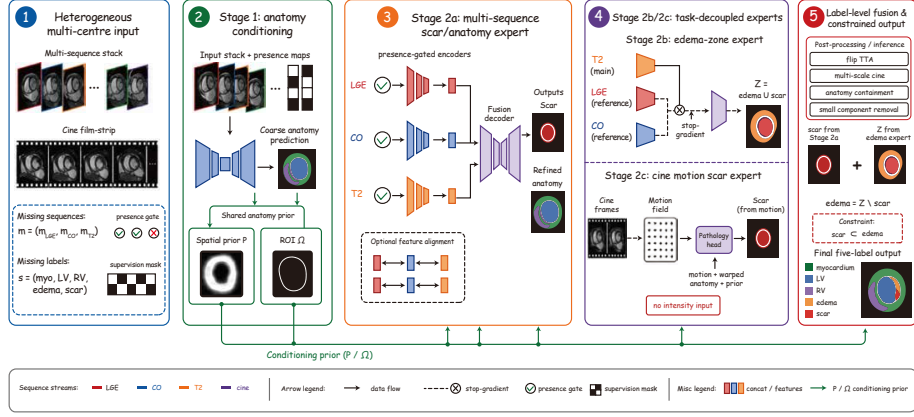


Fig. 1. Overview of MoSAIC. A coarse anatomy stage defines the spatial prior; separate multi-sequence scar, edema-zone, and cine-motion branches produce task-specific predictions that are fused in the challenge label space.

3 Preliminaries

Problem setup. A dataset is a set of tuples $\mathcal{D} = \{(X_i, Y_i, \mathbf{m}_i, \mathbf{s}_i)\}$. The label map $Y \in \{0, \dots, C\}^{D \times H \times W}$ uses the challenge classes {background, myocardium, LV, RV, edema, scar}. For label fusion, we denote the injury zone as $Z = \text{edema} \cup \text{scar}$ and recover pure edema after scar prediction. Two acquisition regimes occur. In the *multi-sequence* regime $X = (\mathbf{I}_{\text{LGE}}, \mathbf{I}_{\text{CO}}, \mathbf{I}_{\text{T2}})$ and $\mathbf{m} \in \{0, 1\}^3$ records which sequences a center actually acquired, with absent sequences zero-filled. In the *cine* regime $X = (I_0, \dots, I_{T-1})$ is a single bSSFP sequence sampled over the cardiac cycle and anchored at end-diastole (ED). In both regimes $\mathbf{s} \in \{0, 1\}^C$ records which classes the originating center annotated; $s_c = 0$ means the label map is silent about class c , leaving its underlying presence unspecified.

Evaluation. Internal cross-validation reports the Dice similarity coefficient and the 95th-percentile Hausdorff distance (HD95, in mm) computed in the resampled physical space. Organizer-scored validation uses the CARE-Myocardium metrics, Dice and Hausdorff distance (HD, in mm).

4 Method

The framework follows a coarse to fine design. Stage 1 performs coarse cardiac anatomy localization and produces a spatial prior. Stage 2 splits fine decoding into a multi-sequence scar/anatomy expert, a dedicated edema-zone expert and a cine motion scar expert. We describe each in turn.

4.1 Stage 1: Anatomy Conditioning

A compact 2D U-Net [19] F_{coarse} predicts three coarse classes (myocardial ring, LV, RV) from the channel-stacked sequences augmented with constant *presence maps*, i.e. the input is $[\mathbf{I}_{\text{LGE}}, \mathbf{I}_{\text{C0}}, \mathbf{I}_{\text{T2}}, m_1 \mathbf{1}, m_2 \mathbf{1}, m_3 \mathbf{1}]$. The presence maps let one network serve every acquisition protocol: the decoder can condition on which channels are informative and avoids interpreting a zero-filled channel as dark tissue. Each output channel is thresholded and reduced to its largest connected component, giving $\hat{Y}^c$ and the binary prior $P = \mathbf{1}[\hat{Y}^c > 0]$. The bounding box of P , dilated by a fixed margin, defines the region of interest Ω passed to Stage 2. Because P is derived from anatomy rather than pathology, it can be computed for cases with or without lesion annotations and is used as the common conditioning signal for the fine stage.

4.2 Stage 2a: Multi-Sequence Scar and Anatomy Decoding

Role. Stage 2a is the multi-sequence scar/anatomy expert. It predicts scar from LGE-centered fused features and uses auxiliary myocardium decoding to keep the fused representation anatomy-aware; edema-zone decoding is handled by Stage 2b.

Modality-decoupled encoding with presence gating. The framework runs three independent encoders $E^{(m)}$, $m \in \{\text{LGE}, \text{C0}, \text{T2}\}$, each producing a four-level pyramid $f_\ell^{(m)}$, $\ell = 1, \dots, 4$. Missing sequences are suppressed multiplicatively,

$$\tilde{f}_\ell^{(m)} = m_m \cdot f_\ell^{(m)}, \quad (1)$$

so an absent auxiliary sequence contributes no image-derived feature maps; the decoder still receives fixed sigmoid gate values from the zeroed features. The same weights serve LGE-only, LGE+C0 and full three-sequence cases.

An optional alignment branch. Separate breath-holds can introduce fine non-rigid offsets around the thin myocardial ring where lesion decisions are made. The framework therefore includes a deformable branch that models sequence alignment inside the representation the decoder consumes: for each moving sequence $m \in \{\text{C0}, \text{T2}\}$ a localization head $h^{(m)}$ reads the *paired* bottlenecks and predicts $K = g^2$ control-point offsets $\theta^{(m)} = h^{(m)}([\tilde{f}_4^{(m)}, f_4^{(\text{LGE})}])$, defining a thin-plate spline [2] $\mathcal{T}_\theta$ that warps every pyramid level of that sequence (Appendix A). The head is zero-initialized for stable alignment learning under the segmentation objective.

Fusion and prior gating. The decoder fuses the sequence pyramids with a gating rule that keeps LGE as the structural carrier while the other sequences modulate it, and the Stage-1 prior re-enters at every decoder level as a multiplicative gate that emphasizes the cardiac region while retaining context. Appendix B gives the equations and the fixed gate values induced by absent auxiliary sequences.

4.3 A Supervision-Masked Objective for Ragged Annotation

Training pools cases with different annotated class subsets by applying the per-class supervision mask $\mathbf{s}$ directly in the loss normalizer. With $\ell_{b,c}^{\text{dice}}$ and $\ell_{b,c}^{\text{bce}}$ the per-sample per-class Dice and binary cross-entropy terms and w_c a class weight,

$$\mathcal{L}_{\text{seg}} = \frac{\sum_{b,c} s_{b,c} w_c [\ell_{b,c}^{\text{dice}} + \ell_{b,c}^{\text{bce}}]}{\sum_{b,c} s_{b,c}}. \quad (2)$$

Each annotated class contributes its supervised gradient; an unannotated class receives zero weight in both the numerator and the denominator. A case from a scar-focused center therefore supervises the scar head and leaves edema supervision to centers that annotated edema.

Small lesions additionally receive a recall-oriented Tversky term [20] with $\alpha < \beta$ and a focal term [13], both restricted to lesion channels and both masked the same way. Deep supervision is applied at two decoder levels.

An auxiliary *cross-sequence myocardium consistency* term completes Stage 2a (Appendix C). It decodes a myocardium mask from each sequence-specific feature stack and regularizes all available pairs with Dice agreement. This ties representations through shared anatomy without adding annotations.

4.4 Stage 2b: A Dedicated Edema Expert

Scar and edema are modeled with dedicated experts because they rest on different physical contrasts, LGE hyper-enhancement versus T2 hyper-intensity, and receive different supervision density: in this cohort scar is annotated on 220 cases and edema on 80. A separate edema expert lets the T2-driven target use a focused decoder and objective.

The framework therefore trains a separate expert. T2 forms the main stream while the LGE and C0 streams are fused element-wise into a *reference* pyramid e_ℓ^{ref} , which enters the decoder *detached*, $d_\ell = \Psi_\ell([\text{up}(d_{\ell+1}), \text{sg}(e_\ell^{\text{ref}}), e_\ell^{\text{T2}}])$. Anatomical context therefore conditions the decision without letting the edema gradient reshape the LGE/C0 features that the scar expert relies on.

The expert regresses the injury zone $Z = \text{edema} \cup \text{scar}$, matching the label representation used for fusion. Pure edema is recovered at fusion time by assigning the predicted scar first and then filling $Z \setminus \widehat{\text{scar}}$ as edema.

4.5 Stage 2c: Motion-Derived Scar from Cine

For cine-only data, the framework replaces the multi-sequence encoder stack with a motion extractor. A shared encoder embeds each of T ED-anchored frames; an anatomy decoder predicts per-frame myocardium and LV; and a motion decoder predicts a dense displacement field between the ED reference I_0 and frame I_t , bounded to keep it physically plausible:

$$\varphi_t = \tau \tanh(D_{\text{mot}}(f_0, f_t)), \quad \varphi_0 = \mathbf{0}, \quad (3)$$

with τ the maximum displacement as a fraction of the field of view. The field warps both the anatomy probability and the frame, $\hat{A}_t = \mathcal{W}(\sigma(A_t), \varphi_t)$ and $\hat{I}_t = \mathcal{W}(I_t, \varphi_t)$.

The pathology head receives only motion, warped anatomy and the anatomical prior:

$$z_t = \Theta\left(\left[\varphi_t, \hat{A}_t, P\right]\right), \quad \hat{Y} = \text{Conv}_{1 \times 1}\left(\frac{1}{T} \sum_t z_t\right). \quad (4)$$

This restricts the cine scar branch to contrast-free temporal and anatomical cues.

Training combines the masked pathology loss with four auxiliary terms: ED-frame anatomy supervision; consistency between each warped anatomy map and the ED prediction; a photometric term matching $\hat{I}_t$ to I_0 ; and an ℓ_1 smoothness penalty on $\nabla\varphi$ for the displacement field.

4.6 Inference

Both regimes share one skeleton: predict anatomy, restrict to the prior, predict pathology with flip test-time augmentation, then apply anatomical containment and minimum lesion-size cleanup. For the multi-sequence regime, scar comes from the Stage-2a scar/anatomy expert, and edema is assigned from $Z \setminus \widehat{\text{scar}}$ under the injury-zone fusion convention. For cine, predictions are additionally averaged over three through-plane resamplings as the deployment-time handling of through-plane resolution variation.

5 Experiments

Data and data-use declaration. All experiments use the CARE-2026 CARE-Myocardium dataset, comprising the MyoPS multi-sequence cohort and the CineMyoPS cine cohort. We use no other data. We gratefully acknowledge the challenge organizers, and cite the sources as required: [28,18,4,3]. Table 1 summarises the heterogeneity that motivates our design. No pre-trained weights are used; every model is trained from scratch.

Implementation. Volumes are resampled to $1.25 \times 1.25 \times 10.0$ mm (multi-sequence) and $1.25 \times 1.25 \times 8.0$ mm (cine), robustly z-scored per volume, and processed slice-wise at 192×192 . Stage-1 uses 24 base channels, Stage-2a 24, the edema expert 64 and the cine model 16, with instance normalization [23]. We train with AdamW [15] under a cosine schedule [14], gradient clipping at 0.1, and a class-quota batch sampler that guarantees a fixed fraction of lesion-bearing slices per batch. The TPS warm-up is $E_w = 10$ epochs. Cross-validation uses five folds stratified by center and by lesion presence, so every fold preserves the cohort’s modality and annotation mix. Because edema is annotated at two centers, edema metrics are computed on those centers’ cases. All runs use seed 3407.

Metrics. Internal five-fold results are reported as Dice and HD95 (mm), with mean $\pm$ standard deviation across folds. Official validation results are organizer-reported Dice and HD (mm).

Table 1. The CARE-2026 CARE-Myocardium training cohort. Sequence availability and annotated classes both vary by center: 116 of 220 multi-sequence cases provide LGE alone, and 80 carry edema labels. The evaluation cohort provides all three sequences, so a model can learn from partial training protocols and exploit complete inputs at test time.

Centre	Cases	Sequences	Annotated classes
<i>MyoPS (multi-sequence)</i>			
A	81	LGE	myo, LV, scar
B	35	LGE+C0+T2	myo, LV, RV, edema, scar
C	45	LGE+C0+T2	myo, LV, RV, edema, scar
E	7	LGE+C0	myo, LV, RV, scar
F	9	LGE+C0	myo, LV, RV, scar
G	8	LGE+C0	myo, LV, RV, scar
H	35	LGE	scar
<i>CineMyoPS (cine only)</i>			
α	40	Cine	myo, LV, scar
β	24	Cine	myo, LV, scar

5.1 Main Results

Multi-sequence segmentation. The upper block of Table 2 reports five-fold performance on the MyoPS cohort. Anatomy shows high Dice: LV reaches 0.866 and RV 0.841, with myocardium at 0.687 for the thin ring-shaped structure. Among pathology classes, scar reaches 0.668 Dice and 11.69 mm HD95. The scar branch uses LGE as its primary contrast. The lesion union reaches 0.645 Dice and is reported in the same injury-zone label space used for fusion; pure edema is reported as a local two-center T2 endpoint in Table 2.

Cine-only segmentation. The cine block shows strong anatomy (LV 0.919, myocardium 0.695) on the two-center CineMyoPS cohort. The motion-based cine branch reaches 0.457 Dice with an HD95 of 19.68 mm using contrast-free temporal input.

Official validation leaderboard. Table 3 reports the best MoSAIC organizer-scored CARE-2026 validation entries for each official target, using the organizer metrics Dice and HD. The best MoSAIC entries reach 0.6965 Dice for MyoPS scar, 0.5983 for MyoPS edema and 0.2069 for CineMyoPS scar. Compared with the nnU-Net comparator, MoSAIC improves MyoPS scar, CineMyoPS scar, and the mean Dice over the three official targets (0.492 to 0.501).

Table 2. Five-fold cross-validated framework performance, mean $\pm$ s.d. across folds. Pure-edema metrics use cases with edema annotations; the remaining metrics use their corresponding annotated cohorts. Cine scar is inferred from motion alone, without any contrast agent.

Structure	Dice $\uparrow$	HD95 (mm) $\downarrow$
<i>MyoPS: multi-sequence (LGE, C0, T2)</i>		
Myocardium	0.687 ± 0.036	8.29 ± 1.61
LV	0.866 ± 0.015	5.87 ± 1.43
RV	0.841 ± 0.022	6.63 ± 1.80
Pure edema	0.349 ± 0.065	20.21 ± 3.49
Scar	0.668 ± 0.022	11.69 ± 1.06
Lesion union	0.645 ± 0.024	15.87 ± 3.13
<i>CineMyoPS: cine only</i>		
Myocardium	0.695 ± 0.024	5.35 ± 0.30
LV	0.919 ± 0.007	3.87 ± 0.55
Scar	0.457 ± 0.038	19.68 ± 3.88

Table 3. Best organizer-scored MoSAIC entries and the nnU-Net comparator on the CARE-2026 validation leaderboards, reported as Dice/HD (mm). For each target, the MoSAIC row is the submitted entry with the highest Dice.

Leaderboard	Target	MoSAIC Dice $\uparrow$	MoSAIC HD $\downarrow$	nnU-Net Dice $\uparrow$	nnU-Net HD $\downarrow$
MyoPS	scar	0.6965	13.7827	0.6258	14.9844
MyoPS	edema	0.5983	26.7067	0.6691	21.0898
CineMyoPS	scar	0.2069	48.7463	0.1816	52.9706

5.2 Discussion

Summary. Coarse anatomy and case-aware supervision provide a common interface for heterogeneous centers. Stage 1 supplies a shared spatial frame, Stage 2a and Stage 2b separate scar and edema according to their evidence sources, and Stage 2c substitutes temporal motion for contrast in cine CMR. In official validation, the strongest MoSAIC entries improve MyoPS scar and CineMyoPS scar relative to the nnU-Net comparator, while the three-target mean Dice is slightly higher.

Limitations. Cine folds are drawn from the same source centers, whereas official validation may change scanner, frame count, temporal resolution and slice thickness; protocol-normalized flow or explicit frame-rate augmentation is therefore a natural extension. For edema, the current fusion predicts $Z = \text{edema} \cup \text{scar}$ and obtains pure edema by subtracting scar at fusion time. Further work will focus on strengthening edema supervision, calibrating edema-zone fusion and adding inter-slice context.

6 Conclusion

We presented a coarse to fine heterogeneity-aware framework for myocardial pathology segmentation with variable sequence availability, heterogeneous labels and contrast-free cine scar evidence. Its coarse to fine cascade first anchors the problem with cardiac anatomy and then decodes pathology inside the anatomical prior. Presence-gated encoders with a supervision-masked loss allow pooled training across heterogeneous sequence and label availability, the cine branch uses temporal motion evidence, and the edema expert predicts the injury-zone label used for fusion. On official validation, the best MoSAIC Dice entries were 0.6965 for MyoPS scar, 0.5983 for MyoPS edema, and 0.2069 for CineMyoPS scar.

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A Thin-Plate Spline Alignment Field

With target control points $\{p_k\}$ on a regular $g \times g$ lattice spanning $[-r, r]^2$ and predicted offsets $\theta_k^{(m)}$, the thin-plate spline [2] is the interpolant of $\{p_k \mapsto \theta_k^{(m)}\}$ with minimal bending energy:

$$\mathcal{T}_\theta(q) = a_0 + Aq + \sum_{k=1}^K w_k U(\|q - p_k\|), \quad U(d) = d^2 \log d, \quad (5)$$

where (w, a_0, A) solve the standard TPS linear system. That system depends only on $\{p_k\}$, which are fixed, so it is inverted once and cached. The final linear layer of the localization head $h^{(m)}$ is zero-initialized with its bias set to the target lattice, giving a stable starting warp that is updated by the task loss.

B Cross-Modality Gated Fusion

The Stage-2a decoder fuses the aligned pyramids at level ℓ as

$$a_\ell = \frac{1}{2} \left[\sigma(\hat{f}_\ell^{(C0)}) + \sigma(\hat{f}_\ell^{(T2)}) \right], \quad (6)$$

$$d_\ell = \Phi_\ell \left(\left[\text{up}(d_{\ell+1}), f_\ell^{(\text{LGE})} \odot a_\ell, \sigma(\hat{f}_\ell^{(C0)}), \sigma(\hat{f}_\ell^{(T2)}) \right] \right), \quad (7)$$

initialized from the concatenated bottlenecks $d_4 = [f_4^{(\text{LGE})}, \hat{f}_4^{(C0)}, \hat{f}_4^{(T2)}]$. When an auxiliary sequence is absent, its masked feature pyramid is zero, so its sigmoid gate channel is fixed at $\sigma(0) = \frac{1}{2}$. This is a consequence of the gate parameterization, not an identity gate; the absent sequence contributes no image-derived feature map. The Stage-1 prior gates every decoder level, $d_\ell \leftarrow d_\ell \odot \sigma(\text{pool}(P))$.

C Auxiliary Training Objectives

Cross-sequence myocardium consistency. Three auxiliary heads $g^{(m)}$ decode a myocardium mask from each sequence’s own features. For the available sequences $A = \{m : m_m = 1\}$, each head is supervised against the myocardium label and every available pair in $\mathcal{Q} = \{(m, m') : m < m', m, m' \in A\}$ is driven into agreement:

$$\mathcal{L}_{\text{cons}} = \frac{\mathcal{L}_{\text{sup}} + \mathcal{L}_{\text{pair}}}{N}, \quad N = |A| + |\mathcal{Q}|. \quad (8)$$

Here $\mathcal{L}_{\text{sup}}$ sums the sequence-specific myocardium Dice terms, and $\mathcal{L}_{\text{pair}}$ sums the pairwise Dice agreement terms.

Stage-2b edema expert. The reference pyramid is formed as

$$e_\ell^{\text{ref}} = \max(E_\ell^{\text{LGE}}([\mathbf{I}_{\text{LGE}}, M]), E_\ell^{\text{C0}}([\mathbf{I}_{\text{C0}}, M])), \quad e_\ell^{\text{T2}} = E_\ell^{\text{T2}}([\mathbf{I}_{\text{T2}}, M]), \quad (9)$$

with M a dilated cardiac mask from Stage 1.